



# Moving Charges and Magnetism

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# Magnetic Field

Magnetic Field is the region around a magnetic material or a moving electric charge within which the force of magnetism acts.

- In 1819, Danish physicist and chemist **Hans Christian Oersted** discovered that an electric current creates a magnetic field around it.
- The symbol for it is  $B$ . SI unit of  $B$  is Newtons per meter per ampere or Teslas. One tesla (1 T) is defined as the field intensity generating one newton (N) of force per ampere (A) of current per meter of conductor.

# Biot-Savart Law

Biot-Savart's law is an equation that gives the magnetic field produced due to a current-carrying segment. This segment is taken as a vector quantity known as the current element.

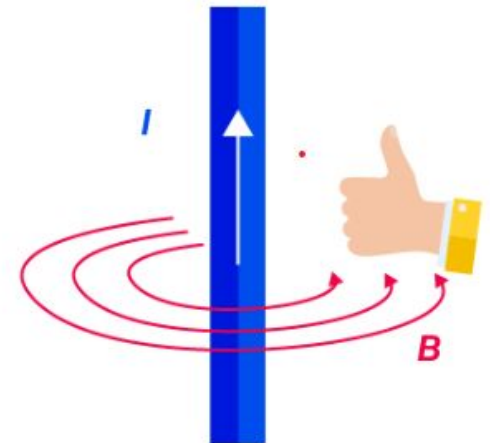
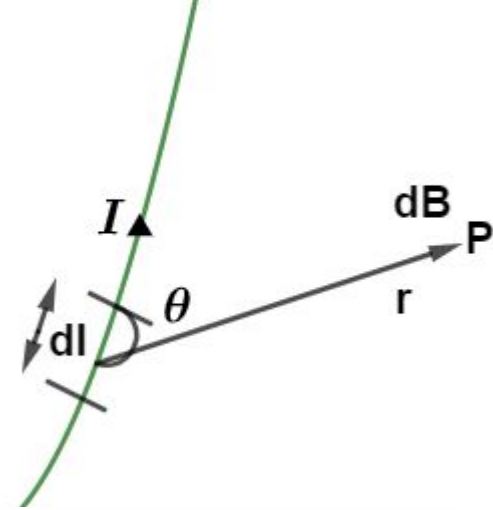
The Biot Savart law is fundamental to magnetostatics, playing a role similar to Coulomb's law in electrostatics.

**Biot-Savart law** was stated by two French physicists, Jean Baptiste Biot and Felix Savart. They derived the mathematical expression for magnetic flux density at a point due to a nearby current-carrying conductor, in 1820.

The magnetic field intensity  $dB$  at a point  $p$  due to this small element  $dl$ , is directly proportional to the length of the element  $dl$ , the current  $I$ , the sine of the angle and  $\theta$  between the direction of the current and the vector joining a given point of the magnetic field and the current element and is inversely proportional to the square of the distance of the given point from the current element,  $r$ .

Hence,  $dB \propto \frac{Idl \sin \theta}{r^2}$  or  $dB = k \frac{Idl \sin \theta}{r^2}$

The direction of the magnetic field is always in a plane perpendicular to the line of element and position vector. It is given by the right-hand thumb rule where the thumb points to the direction of conventional current and the other fingers show the magnetic field's direction.

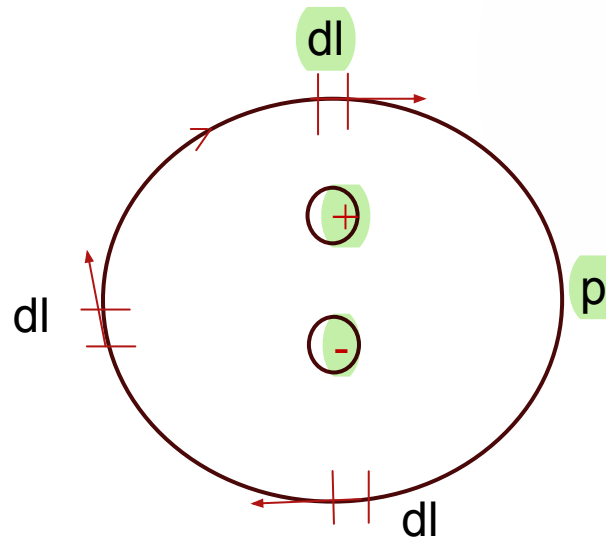


# Ampère's circuital law

Ampère's circuital law relates the integrated magnetic field around a closed loop to the electric current passing through the loop.

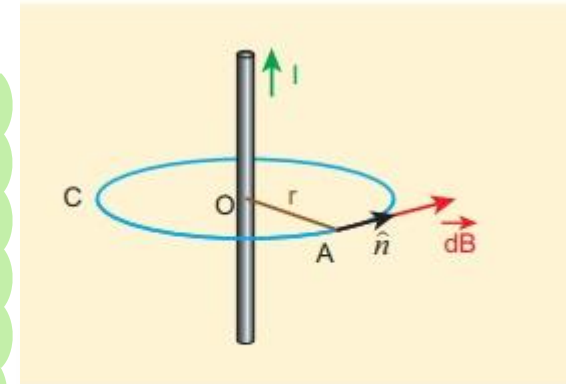
It is an alternate method to find magnetic field strength (the basic method is the Biot Savart Law.)

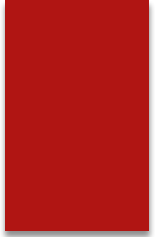
$$\oint B \cdot dl = \mu_0 i_{inside}$$



## Magnetic field due to the current carrying wire of infinite length using Ampère's law

Consider a straight conductor of infinite length carrying current  $I$  and the direction of magnetic field lines is shown in Figure. Since the wire is geometrically cylindrical in shape and symmetrical about its axis, we construct an Ampèrian loop in the form of a circular shape at a distance  $r$  from the center of the conductor as shown in Figure



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1. A car cell phone charger is carrying 50.0 A of current kept at a distance 175 cm away from you. What is the magnetic field in the place you are sitting due to the charger?